The Effects of Occupational Shifts on the Slope of the Phillips Curve: Micro Evidence from Japan

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I. Introduction

The return of inflation following the COVID-19 pandemic has renewed economists' interest in the Phillips curve (Benigno and Eggertsson 2023; Gagliardone and Gertler 2023; Harding et al. 2023). However, whether inflation is there to stay remains an open question. This MPhil thesis attempts to answer this question by looking into the impact of occupational shifts on the slope of the Phillips curve. The main contribution of this study is to connect a strand of the labour economic literature that focuses on the task polarisation of the labour market with a strand of the monetary policy literature related to the inflation puzzle.

Theoretically, I extend a New Keynesian model with search and matching friction (Blanchard and Galí 2010) by adding heterogeneity to the types of workers available to firms. More specifically, I follow the task polarisation literature by recognising that the labour market is segmented along various occupations, such that workers with different profiles are unable to (or unwilling to) perform tasks for which they are not qualified. In addition, I assume each type of worker performs a distinct task for the firm, resulting in a production technology that treats different types of labour inputs as complements. As a result, the production technology shapes the firm's occupational composition, which in turn dictates the firms' exposure to each segment of the labour market. Since each segment is characterised by various degrees of tightness, occupational composition has an effect on the sensitivity of the firm's marginal cost to economic slack.

Empirically, I contribute to the growing body of literature that uses panel data to estimate the slope of the Phillips curve (e.g. Hazell et al. 2020; Imbs et al. 2011; etc...). I collect a novel dataset of 16 industries that are representative of the Japanese manufacturing sector to test the effects of task polarisation on the relationship between prices and economic slack. The results of this exercise indicate that the destruction of routine jobs in favour of non-routine cognitive jobs lead to a flattening of the Phillips curve, as firms' marginal cost becomes less exposed to search cost.

The policy implications of my findings are twofold. Firstly, task polarisation during the COVID-19 pandemic has flattened the Phillips curve in Japan, indicating that other factors might explain the resurgence of inflation in recent years. In fact, my estimates reveal a flatter Phillips curve in the period following the COVID-19 pandemic, such that an aggressive monetary policy stance by the Bank of Japan may not be appropriate in its pursuit of price stability. Secondly and more generally, occupational composition is an important component of firms' price setting decision.

To the extent that the Phillips curve is never perfectly identified, central banks may have to take this factor into account when assessing the inflation-output trade-off they face.

A. Related literature

This MPhil thesis connects two strands of the economic literature. The first strand is related to the so-called "inflation puzzle". A large number of papers have sought to explain the apparent disconnect between inflation and economic slack observed in developed economies since the 1990s. In that regard, two prominent hypotheses have been put forward. One set of explanations is related to the increased efficiency of monetary policy at stabilising prices since the 1980s. The policy hypothesis attributes the low volatility of inflation to the forceful response of central banks to shocks, which has both flattened the slope of the aggregate demand curve (e.g. Barnichon and Mesters 2020; Hooper et al. 2020; McLeay and Tenreyro 2020) and anchored inflation expectation (e.g. Ball and Mazumder 2019; Blanchard 2016; Carvalho et al. 2023). This hypothesis implies that the Philips curve has remained dormant, such that failure by central banks to stabilise prices may have resulted in the return of inflation following the COVID-19 pandemic. To the extent that my thesis focuses on the slope of the Philips curve, the role of monetary policy will be outside the scope of my inquiry. However, as demonstrated in Section IV., policy responses to shocks will pose an identification problem which I address through the estimation of a sectoral Philips curve.

Another set of explanations emphasises changes in the slope of either the price or the wage Philips curve. Studies of the price Phillips curve impute the disconnect between real economic activity and price inflation to a lower pass-through from firms' marginal cost to prices, which could be due to transformations in the goods market (e.g. Forbes 2019; Obstfeld 2019). By contrast, research on the wage Phillips curve attributes this disconnect to structural changes in the labour market and the decreased sensitivity of firms' marginal cost to labour market tightness (e.g. Faccini and Melosi 2021, Hirata et al. 2020, Schmitt-Grohé and Uribe 2022). My thesis belongs to the latter body of literature as it investigates the role of labour market fictions. It contrasts with previous studies however, insofar that the main object of my analysis is the role played by occupational composition in explaining changes in the slope of the Phillips curve, as opposed to labour market flow and downward wage rigidity.

The second strand is about the task-polarisation of the labour market. The

starting point of the task-polarisation model is to conceptualise a production process as a series of distinct tasks to be completed by workers. Acemoglu and Autor (2011) broadly categorise occupations into two dimension, according to the tasks that those occupations entail: "cognitive" versus "manual", and "routine" versus "non-routine". One the one hand, the difference between cognitive and manual jobs is characterised by the degree to which they require the worker to perform a physical activity compared to a mental activity. On the other hand, the distinction between routine and non-routine jobs is defined by how repetitive the tasks performed in those occupations are.

Following this taxonomy, manual occupations correspond to what are popularly described as blue-collar jobs. For instance, the routine manual category include occupations such as packaging worker, construction worker and machine operator, while the non-routine manual category include occupations such as waiters and care workers. The routine cognitive category by contrast, are occupations that are commonly referred to as mid-level white collar jobs such as clerical and sales workers. Finally, the non-routine cognitive category is comprised of professions that require a high level of specialisation such as investment analyst, engineer and lawyer.

In light of this classification, the task-polarisation model postulates that manual jobs, which are more susceptible to automation due to their repetitive nature, have been disappearing since the 1980s in favour of an increase in the share of cognitive non-routine and non-cognitive non-routine jobs (Autor, Katz, and Kearney 2006). Empirically, task polarisation has been widely documented across developed economies and their implication on unemployment and wage inequality are well-known (Acemoglu and Restrepo 2020; Acemoglu 2002; Autor, Katz, and Krueger 1998; etc...). This MPhil thesis attempts to connects this well-documented phenomenon with the widely studied inflation puzzle in monetary policy. In this regard, this paper is similar to Siena and Zago (2024) who also investigate the effects of task polarisation on the slope of the Phillips curve.

Finally, it is noteworthy that my analysis focuses entirely on Japan. While the aim of this thesis is to draw conclusions that are generalisable to other countries, I chose to focus on the Japanese case because it is a canonical example of an economy that has been facing the inflation puzzle since the 1990s. In that respect, my paper will be adding to a large body of existing work covering the relationship between labour market frictions and inflation in Japan (e.g. Fukunaga et al. 2023; Iwasaki et al. 2021; Kubota et al. 2022).

The rest of the paper is organised as follows. In Section II., I first introduce a

structural model to formalise my hypothesis. In Section III., I proceed by presenting the data and a series of key facts which will guide my empirical inquiry. Section IV. specifies the reduced form model and presents the results. Section V. interprets those results and discusses the strengths and limitations of my estimation method. Finally, Section VI. concludes.

II. Theoretical Framework

I begin by providing a theoretical framework to guide my empirical inquiry. I present part of a New Keynesian model with search and matching frictions to derive an expression for the Phillips curve. Since a full exposition of the model is given by Blanchard and Galí (2010), I provide a summary of the derivation with only the key intuition to understand the remainder of this thesis.

A. Assumptions

The representative household is composed of a continuum of members represented by the unit interval $N_t \in [0, 1]$. The household maximises:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left(\log C_t - \chi \frac{N_t^{1+\phi}}{1+\phi} \right) \tag{1}$$

where C_t is a bundle of goods aggregated with a CES function and ϕ is the inverse Frisch labour supply elasticity. Suppose there exists a continuum of firms $i \in [0, 1]$ producing a differentiated final good:

$$Y_t(i) = X_t(i) \tag{2}$$

Intermediate goods producers $j \in [0, 1]$ are in perfect competition, and have access to the identical technology:

$$X_t(j) = A_t N_t \tag{3}$$

where A_t is productivity.

Employment evolves according to the law of motion:

$$N_t(j) = (1 - \delta) N_{t-1}(j) + H_t(j) \tag{4}$$

$H_t(j)$ is a measure of workers firm j decides to hire in period t. In every period, a fraction δ of workers lose their job and join the pool of unemployed workers. Unem-

ployment, denoted U_t, evolves according to:

$$U_t = 1 - N_{t-1} + \delta N_{t-1} = 1 - (1 - \delta)N_{t-1} \tag{5}$$

Aggregating over all intermediate firms, total hires and employment are defined as $H_t \equiv \int_0^1 H_t(j)dj$ and $N_t \equiv \int_0^1 N_t(j)dj$, respectively. Equation (4) can then be rewritten in aggregate form, as:

$$H_t = N_t - (1 - \delta)N_{t-1} \tag{6}$$

I can now define labour market tightness. It is the model's key measure of economic slack, defined as:

$$x_t \equiv \frac{H_t}{N_t} \tag{7}$$

Now, assume hiring is costly. In particular, each intermediate firm faces a cost of G_t for each hire, which is an increasing and concave function of labour market tightness:

$$G_t = A_t B x_t^{\alpha} \tag{8}$$

where $B \in [0, 1]$ is a parameter capturing labour market friction with the resulting cost expressed in terms of the consumption bundle C_t. Here, $B = 0$ would correspond to a frictionless economy, whereas $B > 0$ introduces a search cost to the firm. For compactness, denote $g_t = Bx_t^{\alpha}$. To complete the model, I express this economy's aggregate resource constraint as:

$$C_t = A_t(N_t - Bx_t^{\alpha} H_t) \tag{9}$$

B. Equilibrium conditions

First, suppose firms are price takers. Let P_t be the price of the consumption bundle, $P_t(j)$ be the price set by the producer of the intermediate good j and W_t be the real wage. Intermediate firms maximise:

$$\max_{\{N_t(j), H_t(j)\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t \frac{1}{C_t} \left[\left(\frac{P_t(j)}{P_t} \right) A_t N_t(j) - W_t N_t(j) - G_t H_t(j) \right] \tag{10}$$

subject to equation (4). The first order condition is given by:

$$\left(\frac{P_t(j)}{P_t} \right) A_t = W_t + G_t - \beta(1 - \delta) E_t \left\{ \frac{C_t}{C_{t+1}} G_{t+1} \right\} \tag{11}$$

which equates the firm's marginal benefit of hiring to its marginal cost. In particular, the right-hand-side represents the firm's cost from hiring at t net of the benefits from not paying the hiring cost at $t+1$. Substituting hiring cost (8) into this expression, I obtain:

$$\frac{P_t(j)}{P_t} = \frac{W_t}{A_t} + Bx_t^{\alpha} - \beta(1-\delta)E_t\left\{\frac{C_t}{C_{t+1}}\frac{A_{t+1}}{A_t}B_{t+1}x_{t+1}^{\alpha}\right\} \tag{12}$$

Here, Blanchard and Galí (2010) introduce real wage rigidity. They set $\gamma \in [0,1]$ as an index determining rigidity and Θ as a parameter defined such that the mean wage with real rigidity coincides with its counterpart under Nash bargaining [1]. The resulting expression for real wage is simply written as:

$$W_t = \Theta A_t^{1-\gamma} \tag{13}$$

Replacing W_t in equation (12) and noting that real marginal cost is $MC_t \equiv P_t(j)\,/\,P_t$, the equilibrium condition under real wage rigidity is derived:

$$MC_t = \Theta A_t^{-\gamma} + Bx_t^{\alpha} - \beta(1-\delta)E_t\left\{\frac{C_t}{C_{t+1}}\frac{A_{t+1}}{A_t}B_{t+1}x_{t+1}^{\alpha}\right\} \tag{14}$$

Expression (14) is the key equation in my analysis. It formulates the intermediate firm's marginal cost as a function of labour market tightness and productivity. In the presence of frictions (i.e. $B > 0$), a tighter labour market maps into a higher marginal cost for the firms, who experience an increase in their hiring cost due to more resources put into search. To complete the derivation of the Phillips curve, I introduce nominal rigidity to the model.

In line with most of the literature in monetary policy, I assume a Calvo-style price setting configuration (Calvo 1983). Suppose the fraction of final goods producers that are able to update their prices in period t are restricted to $1-\theta$. Then, the aggregate price index satisfies:

$$P_t = \left((1-\theta)(P_t^*)^{1-\epsilon} + \theta(P_{t-1}^{1-\epsilon})\right)^{\frac{1}{1-\epsilon}} \tag{15}$$

The optimal price P_t^* chosen by firms that are able to reset is pinned down by the following condition derived from the maximisation problem of the intermediate goods producer:

$$E_t\left\{\sum_{k=0}^{\infty}\theta^k Q_{t,t+k}Y_{t+k|t}(P_t^* - \mathcal{M}P_{t+k}MC_{t+k})\right\} = 0 \tag{16}$$

[1] Blanchard and Galí (2010) define $\Theta \equiv (\frac{1}{\mathcal{M}} - (1-\beta(1-\delta))Bx_t^{\alpha}$ where $\mathcal{M}$ is the intermediary firm's mark-up.

where $\mathcal{M}$ is the firm's optimal mark-up. Equations (14), (15) and (16) are sufficient to capture equilibrium inflation dynamics in the presence of labour market frictions.

C. Inflation dynamics near steady state

Let hats denote log deviations from a variable's steady-state value. Equations (15) and (16) can be combined and log-linearised into a standard Phillips curve expressed in terms of marginal cost:

$$\pi_t = \beta E_t \pi_{t+1} + \lambda \hat{mc}_t \tag{17}$$

where $\lambda \equiv (1 - \beta\theta)(1 - \theta)/\theta$. Next, log-linearising expression (14) results in an expression for marginal cost:

$$\hat{mc}_t = \alpha g \mathcal{M} \hat{x}_t - \beta(1-\delta) g \mathcal{M} E_t \Big\{ (\hat{c}_t - \hat{a}_t) - (\hat{c}_{t+1} - \hat{a}_{t+1}) + \alpha \hat{x}_{t+1} \Big\} - \Phi\gamma \hat{a}_t \tag{18}$$

To simplify the formulation above, Blanchard and Galí (2010) make two approximations. Firstly, they assume that hiring costs are small compared to output, such that $\hat{c}_t = \hat{a}_t + \hat{n}_t$ [2]. This allows the substitution $(\hat{c}_t - \hat{a}_t) - (\hat{c}_{t+1} - \hat{a}_{t+1}) = \hat{n}_t - \hat{n}_{t+1}$. Secondly, they impose that the job destruction rate δ be small, making variance in employment $\hat{n}_t - \hat{n}_{t-1}$ negligible compared to labour market tightness $\hat{x}_t$ [3]. Therefore, $\hat{n}_t - \hat{n}_{t-1}$ can be ignored in the second term of equation (18). In addition, noting that if δ is small, $\beta(1-\delta) \approx \beta$, the expression for marginal cost simplifies to:

$$\hat{mc}_t = \alpha g \mathcal{M} (\hat{x} - \beta E_t \hat{x}_{t+1}) - \Phi\gamma \hat{a}_t \tag{19}$$

I can substitute the above into equation (17) to derive inflation as a function of labour market tightness and productivity:

$$\pi_t = \alpha g \mathcal{M} \lambda \hat{x}_t - \lambda \Phi \gamma \sum_{k=0}^{\infty} \beta^k E_t \hat{a}_{t+k} \tag{20}$$

Finally, assuming that productivity follows a stationary AR(1) process whereby $\rho \in [0, 1)$ is the autoregressive coefficient, (20) can be solved forward and the Phillips curve becomes:

$$\pi_t = \alpha g \mathcal{M} \lambda \hat{x}_t - \frac{\lambda \Phi \gamma}{1 - \beta\rho} \hat{a}_t \tag{21}$$

Therefore, inflation increases in labour market tightness and decreases in productivity.

[2] To see why this approximation holds, see the Appendix.

[3] The derivation is available in the Appendix.

Intuitively, higher tightness increases the search cost faced by firms, as labour supply becomes scarcer relative to labour demand. The higher cost translates into higher prices, and its effect is strengthened by the mark-up $\mathcal{M}$ and labour market frictions B. Higher productivity reduces the firm's unit-cost, which reduces its marginal cost and prices.

D. Introducing task polarisation

I can now add task polarisation to this model. For simplicity, suppose the labour market is segmented into high skill and low skill occupations. The two types of workers perform distinct tasks and therefore, are complements. Let's assume that on average, high skill occupations are characterised by high search costs for the firm due to the need to find workers with specialised qualifications, whereas low skill occupations are associated with low search costs because they require little to no qualifications. Their respective search costs are denoted $g_t^h = Bx_t^h$ and $g_t^l = Bx_t^l$, where I set $\alpha = 1$ to keep the analysis simple. Define ω as a parameter capturing the share of high-skill workers in the firm's occupational composition [4] such that:

$$g_t = \omega g_t^h + (1-\omega) g_t^l \tag{22}$$

assuming $x_t = \omega x_t^h + (1-\omega) x_t^l$. In steady-state, $g^h > g^l$ implies $x^h > x^l$ [5] and hence, a higher share of high skill workers ω leads to a higher search cost g.

Looking at equation (21), more high skill workers in steady state would imply a steeper Phillips curve. Intuitively, occupational compositions that place a larger weight on the scarcer type of workers increase the steady-state search cost faced by firms. When a positive demand shock affects the economy, aggregate demand increases and the labour market becomes tighter as firms produce more. In particular, the demand for both types of workers goes up as they are complements in the firm's production function. As a result, the search cost of one type has a spillover effect on the search cost of the other, and the magnitude of this spillover is quantified by the occupational composition ω. Hence, a higher share ω increases the firms' marginal cost whenever the labour market on the whole becomes tighter.

Now, suppose ω_t is time-varying and is subject to shocks. Then, the above toy model illustrates the hypothesised effect of task polarisation on the slope of the Phillips curve. As routine jobs are destroyed for exogenous reasons, the labour market

[4] This leads to the firm's production technology being $X_t = A_t \min[\omega N_t^h, (1-\omega) N_t^l]$.

[5] By extension, this implies $H^h > H^l$ and/or $N^h < N^l$ given the definition $x \equiv \frac{H}{N}$.

polarises into two types of occupations: non-routine cognitive types which correspond to high-skill occupations and non-routine manual types which are equivalent to low-skill jobs. Whether the slope of the Phillips curve flattens or becomes steeper as a result is an empirical question which depends on the relative search cost of both types and the direction that polarisation takes.

III. Data, Definitions and Key Facts

In this section, I first describe a panel dataset that I collected on 16 industries of the Japanese manufacturing sector and define key variables which include 1) producer price index, 2) unit marginal cost, 3) labour market tightness, 4) productivity and 5) an instrument for task polarisation. I proceed by presenting two key facts to motivate my analysis of the data.

A. Data and Definitions

I constructed a panel dataset of 16 subsectors spanning between 2003 Q1 and 2023 Q4, comprising 1080 observations in total. The sample comprises JSIC2 [6] digits industries such as food, electronics and automotive, which are broadly representative of the country's manufacturing sector. While the panel is unbalanced, the sample length seems randomly distributed across industries and panel balance does not seem to be a concern (see Appendix).

The two dependent variables in my analyses are inflation and firms' marginal cost. I define inflation as the year-on-year growth of the producer price index, which is the selling price of domestic producers of their output on the domestic market. Seasonally adjusted data on PPI was collected from the Producer Price Index survey conducted monthly by the Bank of Japan, which I aggregated to a quarterly frequency. Since PPI expresses the evolution of item price, each industry in my dataset was matched with a single item according to its primary activity, as defined by the Japan Standard Industry Classification manual.

To construct a measure of unit marginal cost, I collected data on employees' regular earnings, total number of workers and industrial output expressed in value added. Regular earnings and number of workers were taken from the Monthly Labour Survey, while output was collected from the Index of Industrial Production. Since output indices are only indicative of growth rates, I also collected a cross-section on

[6] JSIC2 digits in the Japan Standard Industry Classification system is equivalent in granularity to NAICS2 digits in the North American Industry Classification System.

sectoral output expressed in levels for the year 2020 to infer the quarterly levels of output for the entire sample period. I combine all of the above variables to construct a measure of unit average cost following Galı et al. (2001), expressed as:

$$MC_{i,t} = \frac{1}{(1-\alpha_i)} \frac{W_{i,t} N_{i,t}}{P_{i,t} Y_{i,t}} \tag{23}$$

where $W_{i,t}$ is the average monthly wage paid to employees working in sector i in quarter t, $N_{i,t}$ is the total number of employees, $P_{i,t}Y_{i,t}$ is output expressed in value added and α_i is the sectoral labour share of income. This formulation of marginal cost follows if one assumes a Cobb-Douglas production function with cost-minimising firms.

To measure labour market tightness, job-to-applicant ratio data from the Employment Referrals for General Workers was collected. The job-to-applicant ratio measures the average number of applicants for each vacancy posted by a firm for a given occupation. To construct the industry average job-to-applicant ratio, I gathered additional data on the total number of vacancies posted by each industry for each occupation. The industry job-to-applicant ratio is then a weighted average:

$$x_{i,t} = \sum_k \frac{V_{k,i,t}}{V_{i,t}} x_{k,i,t} \tag{24}$$

where $x_{k,i,t}$ is the job-to-applicant ratio for occupation k demanded by industry i in quarter t, $V_{k,i,t}$ is the number of vacancies posted by industry i to fill occupation k and $V_{i,t} \equiv \sum_k V_{k,i,t}$ is the total number of vacancies posted by industry i.

Here, I deviate from most of the literature on search and matching because the standard vacancy-to-unemployment ratio fails to capture cross-industry heterogeneity in labour market tightness. In particular, since I assume that the labour market is segmented by skills, industries are competing on multiple segments of the labour market to fill a vacancy and face different levels of tightness depending on their occupational composition. Moreover, this measure takes into account the search intensity of applicants, as workers belonging to the pool of job seekers can apply to multiple positions and increase matching efficiency, a fact that is not captured by the standard vacancy-to-unemployment ratio. As a robustness check, I demonstrate that this novel measure of slackness is tightly correlated with the more traditional vacancy-to-unemployment ratio in Figure 5 of the Appendix.

I compute productivity from the series on output, average labour hours and total number of employees described above. Productivity in this case, is calculated as

Table 1: Mean of key variables

		Key variables:			
	weight	$\bar{\pi}_i$	$\bar{mc}_i$	$\bar{x}_t$	$\bar{a}_t$
Sample average	1.000	0.361	−0.925	0.149	−5.387
Business machinery	0.053	0.362	−1.175	0.223	−5.070
Ceramics	0.024	1.727	−0.422	0.116	−6.881
Chemical	0.151	−0.005	−1.416	0.200	−4.691
Electrical machinery	0.087	−0.516	−0.955	0.232	−5.317
Electronics	0.071	−1.870	−0.852	0.206	−5.430
Fabricated metal	0.048	2.154	−0.326	−0.005	−6.129
Food	0.084	1.346	−0.407	0.111	−6.227
General purp. machinery	0.038	0.947	−0.494	0.224	−5.788
Info. & comm. equipment	0.036	−2.498	−0.718	0.220	−5.407
Non-ferrous metal	0.032	4.342	−1.264	0.136	−5.066
Paper	0.023	1.865	−0.737	0.142	−5.713
Plastic	0.037	0.496	−0.543	0.187	−5.923
Production machinery	0.068	0.915	−0.659	0.231	−5.670
Textile	0.009	1.520	0.352	0.241	−6.981
Transportation equipment	0.228	0.117	−1.216	0.135	−5.079
Wood & lumber	0.006	2.637	−0.084	0.104	−6.499

Note: All variables are expressed in log of means.

output per labour hour:

$$A_{i,t} = \frac{Y_{i,t}}{H_{i,t}N_{i,t}} \tag{25}$$

where $H_{i,t}$ is average labour hours in sector i, which can also be found in the Monthly Labour Survey. Real output, which is denoted $Y_{i,t}$, is found by applying an industry-specific price deflator on output measured in value added.

Finally, I construct my measure of occupational composition as the share of routine jobs. Here, I use employment data from the Labour Force Survey which provides the number of employees in each sector, classified by their occupation. I group these occupations along the "cognitive" vs "manual" and "routine" vs "non-routine" dimensions proposed by Acemoglu and Autor (2011). The same exercise is performed by Furukawa and Toyoda (2013) in their study of the Japanese labour market, allowing me to map US occupation categories directly into their Japanese equivalents and classify them by their skill category. As a result, I obtain the share of routine jobs relative to other occupation categories in a given sector, which I use in my instrument for task polarisation.

Table 2: Correlation of key variables with inflation

	Key variables:		
	$Cor(\pi_{i,t}, mc_{i,t})$	$Cor(\pi_{i,t}, x_{i,t})$	$Cor(\pi_{i,t}, a_{i,t})$
Full sample	0.253	−0.085	−0.167
Business machinery	−0.575	0.350	0.723
Ceramics	0.349	0.185	−0.034
Chemical	0.371	−0.421	−0.065
Electrical machinery	0.615	−0.622	−0.642
Electronics	0.721	−0.017	−0.780
Fabricated metal	0.893	0.327	−0.805
Food	0.711	0.270	−0.584
General purp. machinery	−0.650	−0.615	0.800
Info. & comm. equipment	−0.583	−0.168	0.432
Non-ferrous metal	0.179	0.363	−0.168
Paper	0.275	0.147	0.079
Plastic	−0.006	−0.293	−0.008
Production machinery	0.614	0.449	−0.519
Textile	0.523	0.039	0.170
Transportation equipment	−0.023	−0.361	−0.008
Wood & lumber	0.610	0.387	−0.275

Tables 1 and 2 provide the summary statistics of key variables, with weights constructed based on the average annual output produced by each industry over the entire sample period. Large cross sectional variance can be observed from Table 1, while Table 2 indicate a positive correlation between marginal cost and inflation as well as a negative correlation between productivity and inflation for most industries, as expected. The correlation between inflation and tightness is unexpectedly negative in many industries, indicating the potential existence of confounding factors. In the next subsection, I present the novel dataset further through two key facts which will motivate the empirical analysis.

B. Fact 1: The price and wage Philips curves

The upper panel of Figure 1 shows aggregate PPI inflation [7] plotted against labour market tightness for the period 1990Q1 - 2024Q1. The fitted lines represent a set of naive Philips curves whereby inflation is regressed against tightness without any con-

[7] This is different from the sample average PPI inflation, as not all industries in the Japanese economy are represented in the dataset.

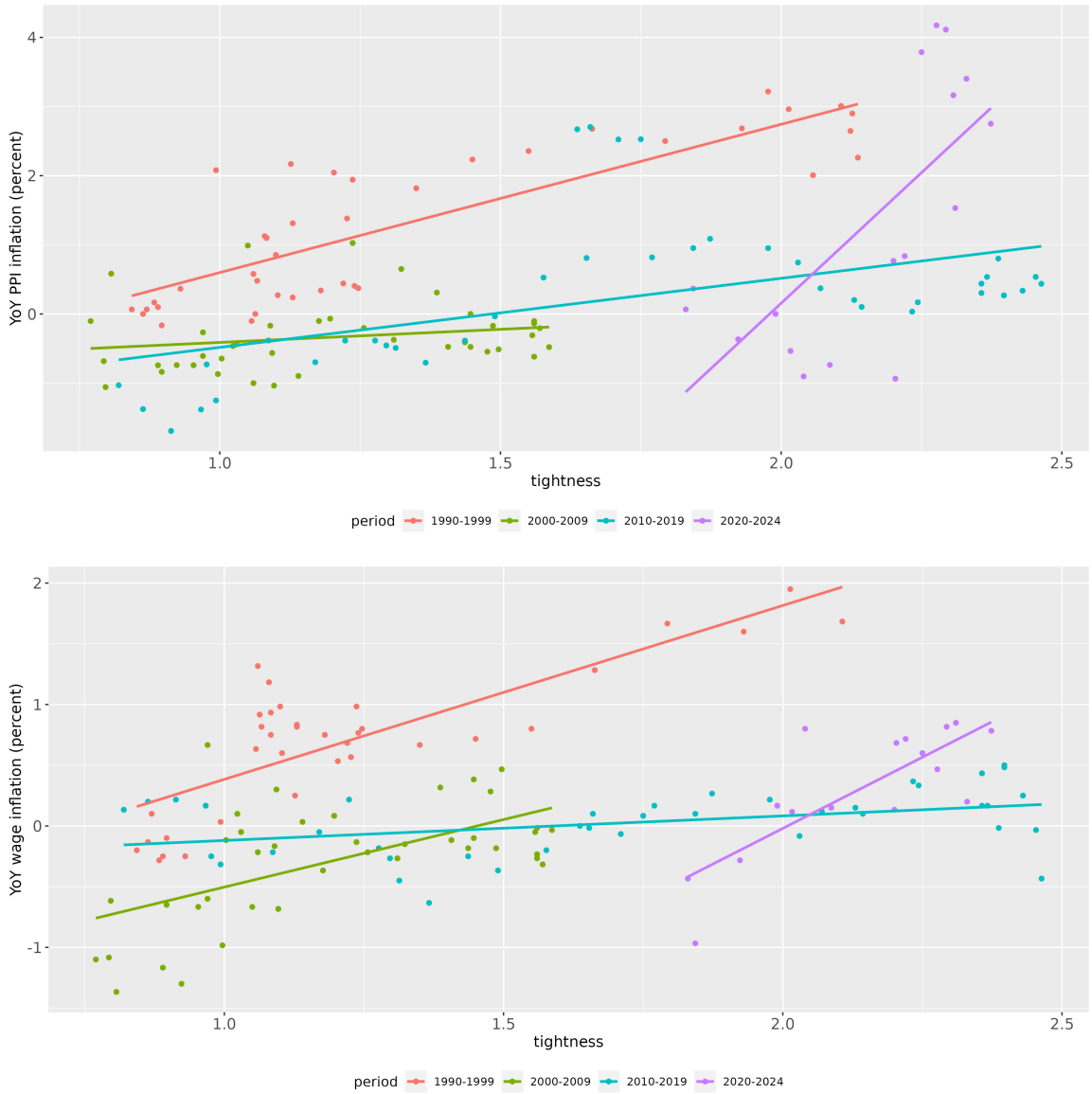


Figure 1: Price and wage Philips curves: 1990-2024

trol variables. I chose to exclude the periods 1970-1979 and 1980-1989 from this analysis, as those periods are characterised by high inflation volatility and un-anchored expectation and hence, a naive estimation yields a negative slope. In line with the pre-existing literature, a large decline the slope of the price Phillips curve can be observed between 2000 and 2019, hinting at a weakening of the relation between labour market tightness and inflation. As documented in recent studies of inflation dynamics in the US (Benigno and Eggertsson 2023), the slope of the Phillips curve in Japan may have also recovered in the period 2020-2024.

The lower panel of Figure 1 shows the nominal wage Phillips curve. Similar to the price Phillips curve, a significant decline in its slope can be observed between the period 1990-2024, hinting that at least some of the changes in the inflation dynamics can be attributed to transformations in the structure of the Japanese labour market. However, upon closer inspection of the wage and price Phillips curves, note that the decline in the former's slope is markedly higher than the decline in the latter's. In fact, while the upper panel demonstrates a recovery of the price Phillips curve between the periods 2000-2009 and 2010-2019, the lower panel shows a weakening of the correlation between wage inflation and economic slack during that same period.

Those results are consistent with previous research on inflation dynamics in Japan (Hoshi and Kashyap 2020; Iwasaki et al. 2021) and in contrast with US studies, which consistently document a sharper decline in the slope of the price Phillips curve compared to the slope of the wage Phillips curve (Del Negro et al. 2020). Consequently, while the flattening of the Phillips curve in the US may be explained by a lower pass-through from marginal cost to prices, the Japanese case may be more indicative of changes in the labour market leading to a lower sensitivity of firms' marginal cost to the country's economic conditions. Therefore, my first hypothesis is that the decline in the slope of the Phillips curve in Japan is mainly driven by structural changes in the labour market.

C. Fact 2: Skill polarisation

The second and perhaps most striking feature of my dataset is the continuous decline in the number of routine workers since the start of the 1990s. In particular, sharp decreases can be observed during periods of recession as shown by the shaded areas in Figure 2, without any recoveries in the subsequent periods. By contrast, the number of non-routine cognitive and non-routine manual workers has been constantly increasing in that same period. Polarisation seems asymmetric during some recessions, with more non-routine manual jobs created during the Great Recession (2008Q1-2009Q1) and more non-routine cognitive jobs created during the COVID-19 pandemic (2018Q4-2020Q4). This form of jobless recovery, characterised by the permanent destruction of routine jobs, is documented both by Jaimovich and Siu (2012) in the US and by Furukawa and Toyoda (2013) in Japan.

While the cyclical nature of occupational composition has been empirically documented in the macro-labour literature, there exists very little theory explaining the lack of recovery in routine occupations following a recession. To the best of my knowledge, Fueki, Katsuki, et al. (2023) offer a possible explanation, albeit in the context

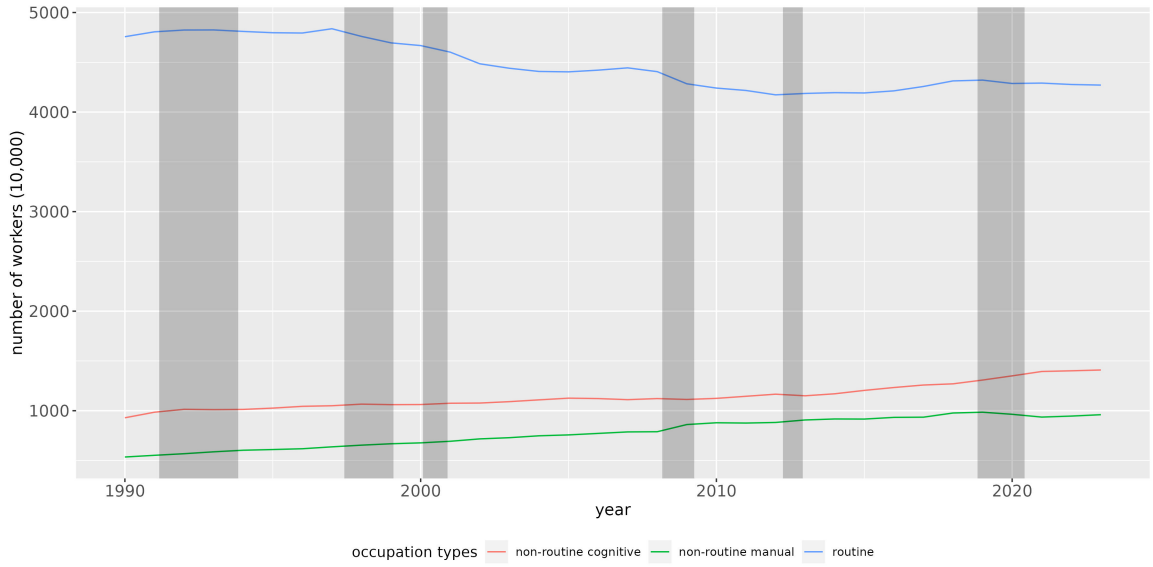


Figure 2: Worker population by skill category: 1990-2024

Note: Shaded areas indicate recessions, as recorded by the Cabinet Office of Japan.

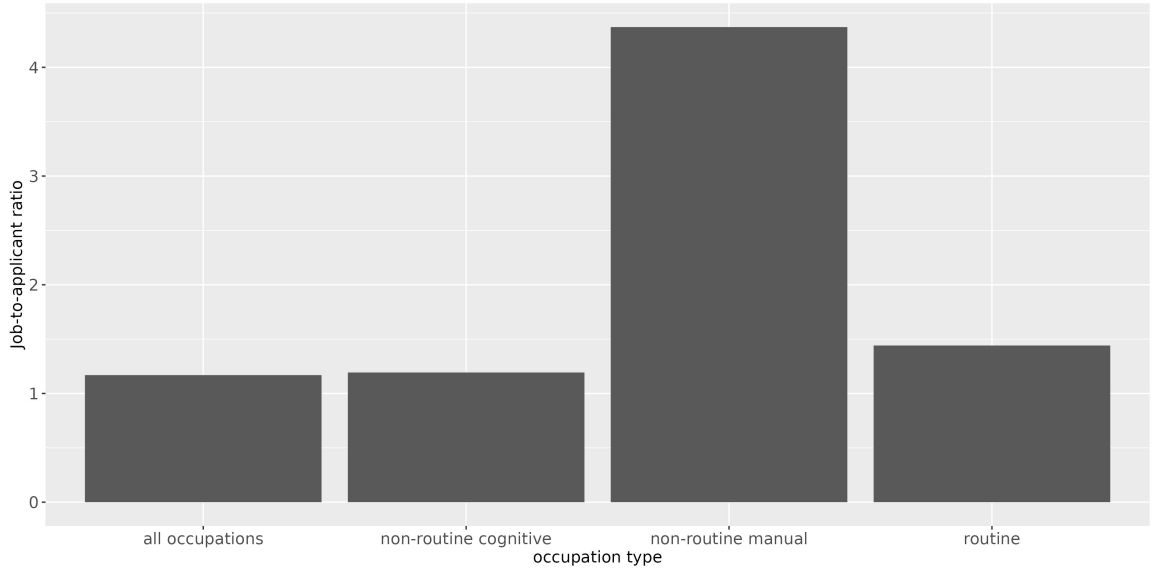


Figure 3: Mean labour market tightness by occupation type

of business cycle and monetary policy. In a New Keynesian framework with automation, they model both labour-augmenting and labour-substituting capital (i.e. robots), with investment in the latter type being endogenous to monetary policy. When the economy is subject to a negative shock, firms reduce their workforce and adjust their production to lower demand. In the ensuing recovery which is characterised by expansionary monetary policy, firms prefer to replace workers for robots as the marginal cost of capital becomes lower relative to the marginal cost of labour.

In the context of task polarisation (which they do not model), this implies routine occupations are substituted by capital.

Hence, two potential effects of skill polarisation on the Phillips curve emerge from this explanation. Firstly, since the share of capital increases relative to the share of labour as automation advances, firms' marginal cost would become increasingly less sensitive to fluctuations in the labour market (Fueki, Maehashi, et al. 2019). Secondly, as the occupational composition polarises towards non-routine cognitive workers with specialised skills (e.g. managers) on the one hand, and towards non-routine manual jobs with unspecialised skills on the other hand (e.g. waiters) at the expense of routine jobs with intermediate levels of specialisation (e.g. accountants, machine operators etc...), firms' search cost on the labour market either increases or decreases, depending on the direction that polarisation takes. To keep my analysis focused on labour frictions, I ignore the automation hypothesis and investigate the task polarisation hypothesis only.

To see that search costs are indeed heterogeneous across occupation types, recall from Section II. the assumption that search cost is an increasing function of tightness: $g_t \equiv Bx_t^{\alpha}$. If search cost varies by occupations, then mean tightness should also differ by occupations, as shown by figure 3. Surprisingly, non-routine cognitive jobs, which are composed of occupations that require highly specialised skills, are characterised by a significantly lower job-to-applicant ratio than non-routine manual jobs. This observation invalidates the intuition formulated in Section II. that firms face a higher cost when searching for high skill workers. This counter-intuitive observation could be driven by the increasing supply of workers with higher education in Japan during the sample period, as well as a shortage of workers with fewer qualifications due to demographic decline (Fukunaga et al. 2023). Nonetheless, this finding should not invalidate the hypothesis that shifts in the occupational composition due to skill polarisation leads to either a flattening or a steepening of the Phillips curve. In light of the evidence presented in Figure 3 and the theoretical framework developed in Section II., polarisation favouring an increase in the share of non-routine manual occupations should steepen the Phillips curve while polarisation favouring an increase in the share of non-routine cognitive occupations should flatten the Phillips curve.

To summarise, two salient facts emerge from this analysis of the aggregate data. Firstly, the decline in the slope of the Phillips curve in Japan appears primarily driven by changes in the structure of the labour market. Secondly, the share of routine jobs has been continuously decreasing since the 1990s, in favour of an increase in non-routine manual and non-routine cognitive occupations. Thus, my hypotheses can be

formally stated as follows:

H1: The decline in the slope of the Phillips curve in Japan during the period 2000-2019 is partially due to occupational shifts.

H2: Task polarisation has an asymmetric effect on the slope of the Phillips curve. When the increase in the share of non-routine cognitive workers dominates, the Phillips curve steepens.

H3: Conversely, when the increase in the share of non-routine manual workers dominates, the Phillips curve flattens.

The next section describes the estimation strategy used to test the above hypotheses.

IV. Estimation of the Phillips Curve

To understand the effects of skill polarisation on the relation between inflation and economic slack, I first need to identify the slope of the Phillips curve. As such, I begin by presenting the simultaneity problem inherent to the estimation of the Phillips curve and how the use of sectoral panel data solves this issue. Next, I define an instrument to measure polarisation, which will be interacted with labour market tightness to understand its effect on the relationship between inflation and economic slack. Finally, I interpret the results through the lens of labour market frictions.

A. Identification Strategy

As an analogy, the identification challenge posed by the Phillips curve is akin to the simultaneity problem an econometrician faces when estimating a good's supply curve (McLeay and Tenreyro 2020). Suppose the central bank sets monetary policy optimally. Then, one can obtain a targeting rule by maximising the policymaker's objective function subject to the Phillips curve. The targeting rule expresses the monetary authority's desired output gap given a level of inflation (or vice-versa), while the Phillips curve represents the private sector's price-setting behaviour given the output gap. Together, they should determine the equilibrium inflation and output, the same way demand and supply curves jointly give an equilibrium price-quantity pair. Then, the challenge when estimating the Phillips curve is to eliminate the resulting simultaneity bias by controlling for the endogenous response of monetary policy to demand shocks.

Some common approaches to the simultaneity problem are controlling for supply shocks, using an exogenous instrument for the output gap and disaggregating the Phillips curve (McLeay and Tenreyro 2020). I use the last approach since the panel structure of my dataset allows me to disaggregate the Phillips curve at the industry level. The model specification broadly follows previous estimations of regional (Hazell et al. 2020) and sectoral Phillips curves (Imbs et al. 2011; Gagliardone, Gertler, et al. 2023), which can be expressed as:

$$\hat{\pi}_{i,t} = \gamma_1 \hat{mc}_{i,t} + \gamma_2 E_t \hat{\pi}_{i,t+1} + \hat{u}_{i,t} \tag{26}$$

where hats denote log-deviation from the aggregate, $\pi_{i,t}$ expresses inflation and $mc_{i,t}$ is marginal cost for industry i in quarter t, as defined in Section III.. I add an industry fixed-effect to control for cross-sectional heterogeneity, which includes differences in the share of labour, dependence on energy import and other time-invariant confounding factors:

$$\hat{\pi}_{i,t} = \alpha_i + \gamma_1 \hat{mc}_{i,t} + \gamma_2 E_t \hat{\pi}_{i,t+1} + \hat{u}_{i,t} \tag{27}$$

Next, note that series on sector-level inflation expectation are available thanks to the quarterly Tankan survey conducted by the Bank of Japan. However, the sample is limited to the period 2014 Q1 - 2023 Q4 and including this variable in the model would significantly reduce the number of observations. As noted by McLeay and Tenreyro (2020) however, it is possible to assume that inflation expectation is homogeneous across sectors if its cross-sectional variance is constant over time, allowing me to substitute the expectation term with a time fixed effect. To check whether this assumption holds, I use the industry-level expectation data to estimate the cross-sectional variance over a limited period [8]. The results indicate time-varying volatility in inflation expectation, which leads me to reject this assumption. Nonetheless, I decided to replace expectation by a time fixed effect due to the prohibitively large reduction in sample size that I would otherwise incur. The resulting model specification is:

$$\pi_{i,t} = \alpha_i + \gamma_1 mc_{i,t} + \delta_t + u_{i,t} \tag{28}$$

where δ_t is a time fixed-effect which controls for expectation. This formulation also allows me to dispense with log-deviations, since the aggregate trends are captured by the time fixed effect. All variables therefore, are now expressed in logs.

[8] The cross-sectional variance in expectation is presented in the Appendix.

To express the Phillips curve in terms of economic slack, an expression for marginal cost needs to be derived. Unlike a New Keynesian Phillips curve with a frictionless labour market, marginal cost is in this case a function of both the output gap and productivity, as derived in equation (19):

$$\hat{mc}_{i,t} = \lambda_1 \hat{x}_{i,t} + \lambda_2 E_t \hat{x}_{i,t+1} + \lambda_3 \hat{a}_{i,t} + \hat{\epsilon}_{i,t} \tag{29}$$

where $x_{i,t}$ is labour market tightness faced by industry i in quarter t as defined in Section III. and $a_{i,t}$ is sectoral productivity. Substituting (29) into (28) and assuming that $E_t \hat{x}_{i,t+1} = E_t \hat{x}_{t+1}$, I obtain the baseline specification which resembles equation (21):

$$\pi_{i,t} = \alpha_i + \beta_1 x_{i,t} + \beta_2 a_{i,t} + \delta_t + u_{i,t} \tag{30}$$

Here, note that the time fixed-effect not only controls expectation but also trends in labour supply that are common across sectors. Given the influx of women and elderly people joining the labour force in Japan to work on part-time contracts during the sample period, it is particularly important to control for this increase in workers as it would both create slackness in the labour market and decrease wages (Fukunaga et al. 2023). Moreover, the time fixed-effect controls for other factors that might affect firms' prices and marginal cost, such as fluctuations in energy prices.

Aggregate inflation is simply a weighted average of sectoral inflation:

$$\pi_t = \sum_i \omega_{i,t} \pi_{i,t} \tag{31}$$

while similarly, aggregate labour market tightness is obtained as:

$$x_t = \sum_i \omega_{i,t} \hat{x}_{i,t} \tag{32}$$

where the weights are measured as the relative output of each industry to the sample total and are defined such that $\Sigma_i \omega_{i,t} = 1$.

To understand why this specification addresses the simultaneity bias, note that the central bank reacts to aggregate demand shock and by extension to x_t, which measures aggregate economic slack. However, it does not react to $\hat{x}_{i,t}$, which is the deviation in labour market tightness idiosyncratic to each industry. As a result, $x_{i,t}$ is uncorrelated with the error $u_{i,t}$ and it is exogenous.

B. Instrument

To relate task polarisation to the slope of the Phillips curve, a measure of occupational shift needs to be constructed. However, using the share of routine jobs directly may pose a problem for at least two reasons. Firstly, the share of routine labour, inflation and tightness may all share a common deterministic trend, leading to spurious correlation. Secondly, occupational shifts follow the business cycle, as discussed in Section III.. This is particularly problematic if one accepts the explanation by Fueki, Katsuki, et al. (2023) that skill polarisation happens when the marginal cost of labour is high relative to the marginal cost of capital, as such periods coincide with expansionary phases in the business cycle when the labour market is tight and inflation is high. The resulting correlation between occupational shift and labour market conditions on the one hand, and between occupational shift and inflation on the other hand, would create an endogeneity bias.

To address this issue, I construct an exogenous instrument by exploiting the dynamics of jobless recoveries documented in Section III.. I follow Siena and Zago (2024) who measure the difference between the peak and the trough share of routine jobs during a recession. Formally, the instrument is defined as:

$$Shift_{i,c} = \frac{Share_{i,c}^{peak} - Share_{i,c}^{trough}}{Share_{i,c}^{trough}} \tag{33}$$

where $Share_{i,c}^{peak}$ and $Share_{i,c}^{trough}$ are the shares of routine jobs during the peak and trough of recession c respectively, in industry i. I refer to the official business cycle dating published by the Japanese Cabinet Office to identify periods of economic contraction, which consist of three episodes in the sample period: the Great Financial Crisis (2008Q1-2009Q1), the recession following the 2011 earthquake and tsunami (2012Q1-2012Q4) and the COVID-19 crisis (2018Q4-2020Q4).

Figure 4 shows the occupational shifts by industries, in which a positive value represents a reduction in the share of routine jobs while a negative value expresses an increase. Across industries, the largest shifts occur during the COVID-19 pandemic, while the effects of the Great Financial Crisis on skill polarisation is ambiguous. The 2011 recession clearly increased the share of routine workers in a handful of industries, leading me to exclude this instrument from the analysis.

Both instruments are exogenous under the assumption that the large occupational shifts that occurred during the Great Financial Crisis and the COVID-19 recession were caused by demand shocks and therefore, were independent from labour market

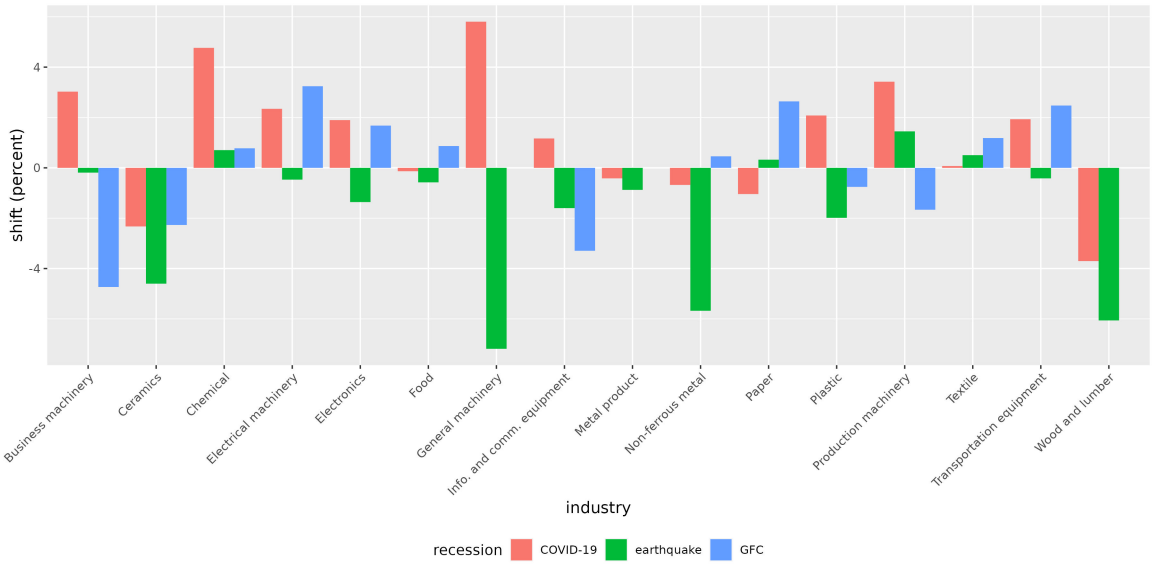


Figure 4: Occupational shifts by industry

dynamics. Then, those instruments can be added to the baseline model to measure the effects of task polarisation on the Phillips curve:

$$\begin{aligned}\pi_{i,t} &= \alpha_i + \kappa x_{i,t} + \gamma_1 a_{i,t} + \delta_t \\ &+ \sum_{c\in\{GFC,COVID\}} \left\{\phi_{1,c} After_{i,c} + \kappa_{1,c} After_{i,c} \times x_{i,t}\right\} \\ &+ \sum_{c\in\{GFC,COVID\}} \left\{\phi_{1,c} After_{i,c} \times Shift_{i,c} + \kappa_{2,c} After_{i,c} \times Shift_{i,c} \times x_{i,t}\right\} \\ &+ u_{i,t}\end{aligned} \tag{34}$$

where $After_{i,c}$ is a dummy variable taking the value of one in all quarters after the recession c. The parameter $\kappa_{2,c}$ captures the causal effect of task polarisation on the slope of the Phillips curve by exploiting the cross-sectional heterogeneity in the magnitude of occupational shifts. It is a proxy for parameter g which measures search cost in the structural equation (21)

C. Results

Table 3 shows the regression output of the baseline Phillips curve for various sample periods. As expected, the coefficients on tightness are positive, those on productivity are negative and the Phillips curve flattens over time, although the increase in its slope between 2010-2019 and 2020-2024 cannot be observed. This might indicate that the resurgence of inflation after the COVID-19 pandemic is due to factors other

than structural changes in the labour market. An important caveat of this analysis however, is the low explanatory power of model (2) to (4) due to the short sample periods and the lack of control variables besides the time and sector fixed effects.

Table 3: Estimation of the baseline Phillips curve

	Dependent variable:			
	inflation			
	(1)	(2)	(3)	(4)
log(tightness)	0.182^{***}	0.097^{***}	0.035^{**}	0.060
	(0.013)	(0.030)	(0.017)	(0.043)
log(productivity)	-0.224^{***}	-0.311^{***}	-0.140^{***}	-0.112^{**}
	(0.020)	(0.096)	(0.022)	(0.047)
time FE	YES	YES	YES	YES
sector FE	YES	YES	YES	YES
sample	full	2000-2009	2010-2019	2020-2024
Observations	1,080	212	612	256
R^2	0.203	0.019	0.047	0.059
Adjusted R^2	0.122	−0.190	−0.049	−0.076
F Statistic	155.544^{***}	6.896^{***}	21.041^{***}	2.972^{*}
Note:				*p<0.1; **p<0.05; ***p<0.01

It is difficult to compare the magnitude of my results with those of other studies, as papers estimating a sectoral Phillips curve regress inflation on marginal cost instead labour market tightness. As an auxiliary test, Table 6 in the Appendix shows inflation regressed against marginal cost under various specifications to evaluate the order of magnitude of parameter γ_1 in equation (28). My estimates of parameter γ_1 range between 0.09 and 0.17 when statistically significant, which is below the range of 0.29 to 0.33 found by Imbs et al. (2011) in their estimation of the French Phillips curve but comparable to the range of 0.05 to 0.6 found by Hazell et al. (2020) in their work on the Belgian Phillips curve.

Table 4 shows an estimation of equation (34). The negative effect of occupational shifts on the slope of the Phillips curve using the GFC instrument is clear, with a reported decline of 0.017 percentage point for every percentage point shift in the share of routine jobs. The COVID instrument also produces a negative estimate, which is coherent with the flattening of the Phillips curve found in the baseline model but

Table 4: The effects of occupational shifts on the Phillips curve

	Dependent variable:	
	inflation	marginal cost
	(1)	(2)
log(productivity)	-0.173^{***}	-0.987^{***}
	(0.024)	(0.013)
log(tightness)	0.078^{***}	-0.009
	(0.010)	(0.005)
log(tightness) × post GFC	-0.086^{*}	0.029
	(0.048)	(0.025)
log(tightness) × shift GFC	-0.024	0.162^{***}
	(0.051)	(0.027)
post GFC × shift GFC	-0.026^{***}	0.001
	(0.007)	(0.004)
log(tightness) × post COVID	0.017^{***}	-0.011^{***}
	(0.006)	(0.003)
log(tightness) × shift COVID	-0.117^{***}	0.038^{**}
	(0.035)	(0.019)
post GFC x shift COVID	-0.019^{**}	0.016^{***}
	(0.008)	(0.004)
log(tightness) × post GFC × shift GFC	-0.016^{***}	-0.002
	(0.003)	(0.002)
log(tightness) × post COVID × shift COVID	0.208^{***}	0.045^{***}
	(0.028)	(0.015)
log(tightness) × post GFC × shift GFC	-0.017^{***}	-0.007^{**}
	(0.005)	(0.003)
log(tightness) × post COVID × shift COVID	-0.037^{***}	-0.008^{***}
	(0.004)	(0.002)
time FE	YES	YES
sector FE	YES	YES
Observations	1,080	1,080
R^2 24	0.495	0.886
Adjusted R^2	0.482	0.883
F Statistic	85.825^{***}	679.462^{***}
Note:	*p<0.1; **p<0.05; ***p<0.01	

at odds with the key facts discussed in Section III. which documents a steepening of the wage Phillips curve after the pandemic. As a robustness check, I re-estimate equation (34) with marginal cost as the dependent variable, to ensure that my results are indeed driven by changes in the structure of the labour market. The GFC and COVID instruments both produce estimates that are negative and statistically significant, which demonstrates that at least some of the effects of task polarisation on price are mediated by firms' marginal cost. Taken together, the results indicate that occupational shifts decreased the slope of the Phillips curve during the Great Recession and during the COVID-19 pandemic.

D. Heterogeneity

While heterogeneity is mainly used for identification, the panel structure of my dataset also allows the analysis of labour market frictions and price setting at an industry level. In particular, my results raise an important question that could be answered by disaggregation: do sectors that are heavily reliant on routine labour experience a larger decline in the slope of their Phillips curve?

To answer this question, I estimate the disaggregated sectoral Phillips curve for each sector as:

$$\pi_t = \alpha + \gamma_1 a_t + \gamma_2 E_t \pi_{t+1} + \kappa_1 x_t + \kappa_2 x_t \times After_{COVID} + u_t \tag{35}$$

where $E_t\pi_{t+1}$ needs to be included as I no longer work with a panel structure. As a result, the sample size is restricted to quarters posterior to 2014Q1, such that only the COVID shock can be used as an instrument. Including expectation also decreases the number of industries for which data is available to 11. Moreover, $Shift_{i,c}$ is dropped from the equation since it is now perfectly colinear with $After_{COVID}$, and the analysis no longer captures the magnitude of the occupational shift as a result. The regression outputs are presented in Table 5 with κ_2 as the key parameter.

The last column shows that the Phillips curve flattened in all industries during the COVID-19 pandemic, except in paper manufacturing. Note however, that paper increased its share of routine jobs during the pandemic, such that a steeper Phillips curve would be expected if the effect of task polarisation is symmetric.

I examine the share of routine jobs in each industry at the beginning of the sample period to see whether those that are reliant on routine occupations experienced more flattening (see Figure 7 in the Appendix). There is no discernible pattern in the regression results, as some industries with a high share of routine labour (e.g. textile)

Table 5: Estimation of the sectoral Phillips curves

	Parameters:			
	γ_1	γ_2	κ_1	κ_2
Business machinery	0.005	0.009^{***}	0.017^{*}	-0.046^{**}
	(0.042)	(0.003)	(0.009)	(0.022)
Ceramics	-0.390^{**}	0.027^{***}	0.022	-0.279^{**}
	(0.152)	(0.005)	(0.058)	(0.118)
Chemical	−0.015	0.067^{***}	−0.029	−0.014
	(0.122)	(0.009)	(0.032)	(0.078)
Electrical machinery	−0.043	0.042^{***}	-0.035^{**}	-0.126^{*}
	(0.072)	(0.004)	(0.015)	(0.038)
Food	-0.180^{*}	0.019^{***}	−0.010	0.001
	(0.101)	(0.005)	(0.101)	(0.005)
Info. & comm. equipment	0.000	0.039^{***}	−0.015	-0.068^{***}
	(0.018)	(0.004)	(0.010)	(0.024)
Non-ferrous metal	0.013	0.080^{***}	−0.054	0.144
	(0.294)	(0.014)	(0.050)	(0.126)
Paper	-0.478^{**}	0.003	0.018	0.252^{**}
	(0.188)	(0.005)	(0.049)	(0.115)
Production machinery	0.036	0.021^{***}	-0.032^{**}	-0.079^{**}
	(0.056)	(0.003)	(0.013)	(0.035)
Textile	-0.231^{***}	0.027^{***}	0.025^{*}	-0.093^{*}
	(0.049)	(0.004)	(0.014)	(0.049)
Transportation equipment	0.050^{**}	0.025^{***}	-0.026^{***}	0.008
	(0.021)	(0.003)	(0.007)	(0.028)
Note:				*p<0.1; **p<0.05; ***p<0.01

experienced a large decrease in the slope of their Phillips curve, while other industries with a small share of routine labour (e.g. electrical machinery) were also subject to some equally important drops in their slopes. Consequently, a larger and more

representative sample of Japanese industries may be needed to conclusively state whether the flattening of the Phillips curve is driven by routine-intensive sectors.

V. Discussions

The results show a clear effect of occupational shifts on the slope of the Phillips curve using both the GFC and COVID instrument. Therefore, the above empirical exercise validates Hypothesis 1 that skill polarisation resulting from the destruction of routine jobs affects the slope of the Phillips curve. Given that the occupational composition of all industries in my sample consist of a mixture of only routine workers and non-routine cognitive workers, the flattening of the Phillips curve found in my model also provides evidence for Hypothesis 2. Since no industry in my sample employs a significant number of non-routine manual workers, it is impossible to test Hypothesis 3.

In addition to previous studies (Fukunaga et al. 2023; Iwasaki et al. 2021), these results provide further evidence that changes in the structure of the labour market are responsible for the flattening of the Phillips curve in Japan. In particular, my results show that shifts in occupational composition has an effect on the slope of the Phillips curve through changes in the search cost faced by firms, as the labour market is segmented by skills. This story contrasts with Siena and Zago (2024) who also investigate the impact of occupational shifts, as their paper finds that differences in the job finding rate and job destruction rate (i.e. "labour market fluidity") between occupations explains the flattening of the Phillips curve. In that sense, their story is closer to a dual labour market hypothesis which claims a labour market that is strongly segmented along different levels of employment protection results in a more sclerotic labour market and a flatter Phillips curve.

From a policy standpoint, my findings imply factors other than a dual labour market can explain the flattening of the Phillips curve, in a way that labour market reforms may not entirely bring the Phillips curve back to life. More specifically, insofar as workers are heterogeneous along their skills and the tasks they perform, the labour market is segmented in a way that changes in occupational composition in itself has an effect on the slope of the Phillips curve, as the degree of friction varies from one occupational category to another. Hence, misallocation of resources such as an over-supply of workers with higher education or a shortage of workers willing to perform a manual occupation may have an effect on the sensitivity of firms' marginal cost to economic slack due to differences in search costs. Moreover, to the extent that factors

such as automation and foreign outsourcing affect the occupational composition of the workforce (Autor, Dorn, et al. 2015), central banks need to take into account such shocks when attempting to identify the evolution in the inflation-output gap trade-off they face.

Methodologically, this thesis contributes to a growing body of studies using panel data to identify the slope of the Phillips curve (e.g. Hazell et al. 2020; Gagliardone, Gertler, et al. 2023; Imbs et al. 2011; etc...). The main advantage of this method is the depth of analysis that the granularity of the data allows, as I am able to dissect the Phillips curve and understand the transmission mechanism from labour market tightness to marginal cost, and from marginal cost to prices. Moreover, despite the limited success in my interpretation of heterogeneity, panel methods offer the possibility to remove confounding factors by sectioning the sample along relevant dimensions (e.g. tradeable vs non-tradeable), an exercise which is possible if the dataset can be expanded beyond the manufacturing sector. Given the richness of Japanese sectoral data and the potential existence of similar data from other countries, expanding this analysis to a larger dataset could indeed be an avenue for future research which would add robustness to my results.

From a theoretical perspective, an important question raised by this study is whether monetary policy has a feedback effect on skill polarisation. Clearly, the simple model presented by Blanchard and Galí (2010) is not satisfactory in capturing the full depth of this problem. Endogenising occupational composition in a New Keynesian model with search and matching friction may help elucidate this question. Moreover, the search cost function $g_t \equiv Bx_t^{\alpha}$ as specified by Blanchard and Galí (2010) needs to be micro-founded, as the interpretation for the friction parameter B is unclear.

VI. Conclusion

The resurgence of inflation in developed economies after the COVID-19 pandemic has brought the Phillips curve back to the forefront of economists' research agenda. Whether this inflationary episode is the pure product of a temporary shock or whether it is explained by a permanent transformation in the structure of the labour market remains an open question. This thesis attempts to provide an answer by elucidating the relationship between the flattening of the Phillips curve and the decline in the share of routine jobs, both observed since the 1990s.

I begin by extending a New Keynesian model with search and matching frictions

developed by Blanchard and Galí (2010) to provide a mechanism that explains how occupational composition affects firms' pricing behaviour. More specifically, I assume that various types of workers perform distinct tasks for the firm, making them complementary with each other. Then, it follows that the cost of hiring one type of worker has a spillover effect on the cost of hiring another type, and the extent of this effect is determined by the occupational composition of firms. As a result, a shift in the share of one type affects the sensitivity of firms' marginal cost to labour market tightness and therefore, it influences the slope of the Phillips curve.

To uncover the causal effect of task polarisation on inflation dynamics, I collect a unique panel dataset of Japanese industries spanning two decades. Using the acceleration of polarisation during recessions as an external instrument, I demonstrate that occupational shift has indeed an impact on the relationship between inflation and economic slack. Since all industries in the sample employ a mixture of either routine workers or non-routine cognitive workers, only the effects of task polarisation that favours the latter type could be investigated. With that respect, my model is successful in uncovering the negative effect of an increase in the share of non-routine cognitive workers on the slope of the Phillips curve but fails to test the impact of an increase in the share of non-routine manual workers. Extending the sample to more industries could overcome this limitation, as non-routine manual occupations (e.g. waiter, cleaning workers, healthcare workers etc...) are more closely associated with the service sector.

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VII. Appendix

A. Derivation of the approximations

The aim is to simplify the following expression:

$$\hat{mc}_t = \alpha g \mathcal{M} \hat{x}_t - \beta(1-\delta) g \mathcal{M} E_t \Big\{ (\hat{c}_t - \hat{a}_t) - (\hat{c}_{t+1} - \hat{a}_{t+1}) + \alpha \hat{x}_{t+1} \Big\} - \Phi \gamma \hat{a}_t \tag{36}$$

$\hat{c}_t = \hat{a}_t + \hat{n}_t$ if the hiring cost g is small relative to output. To see why, recall the aggregate resource constraint, which is one of the equilibrium conditions:

$$C_t = A_t (N_t - B x_t^{\alpha} H_t) \tag{37}$$

which is log-linearised as:

$$\hat{c}_t = \hat{a}_t + \frac{1-g}{1-\delta g} \hat{n}_t + \frac{g(1-\delta)}{1-\delta g} \hat{n}_{t-1} - \frac{\alpha g}{1-\delta g} \delta \hat{x}_t \tag{38}$$

Now, suppose $g \approx 0$. Then, it follows immediately that $\hat{c}_t = \hat{a}_t + \hat{n}_t$ holds. Substitute this expression into equation 36 to get:

$$\hat{mc}_t = \alpha g \mathcal{M} \hat{x}_t - \beta(1-\delta) g \mathcal{M} E_t \Big\{ (\hat{n}_t - \hat{n}_{t+1}) + \alpha \hat{x}_{t+1} \Big\} - \Phi \gamma \hat{a}_t \tag{39}$$

Next, to understand why the approximation $\hat{n}_t - \hat{n}_{t+1} \approx 0$ holds, I start from the definition of labour market tightness:

$$x_t = \frac{H_t}{U_t} \tag{40}$$

Substituting the expressions for unemployment and hiring, I can express tightness in terms of employment only:

$$x_t = \frac{1 - (1-\delta) N_{t-1}}{N_t - (1-\delta) N_{t-1}} \tag{41}$$

The above formulation is log-linearised as:

$$\delta \hat{x}_t = \hat{n}_t - (1-\delta)(1-x) \hat{n}_{t-1} \tag{42}$$

A small δ implies $\hat{x}_t$ needs to be large relative to $\hat{n}_t - \hat{n}_{t+1}$ for the above equality to hold. Turning back to equation 39, note that the term inside the expectation operator is a linear combination of $\hat{n}_t - \hat{n}_{t+1}$ and $\hat{x}_{t+1}$. If $\hat{n}_t - \hat{n}_{t+1}$ is significantly smaller than

$\hat{x}_t$, then it can be ignored completely and this yields:

$$\hat{mc}_t = \alpha g \mathcal{M} \hat{x}_t - \beta(1-\delta) g \mathcal{M} E_t\left\{\alpha \hat{x}_{t+1}\right\} - \Phi\gamma\hat{a}_t \tag{43}$$

B. Additional descriptions of the data

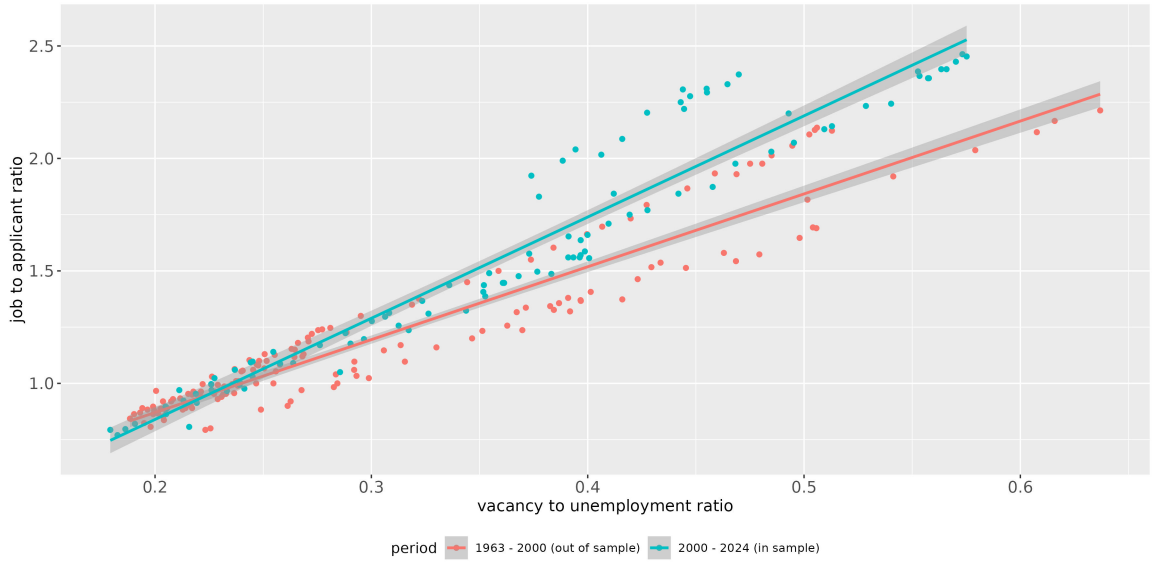


Figure 5: Correlation between vacancy-unemployment ratio and job-applicant ratio

C. Auxiliary results to Section IV

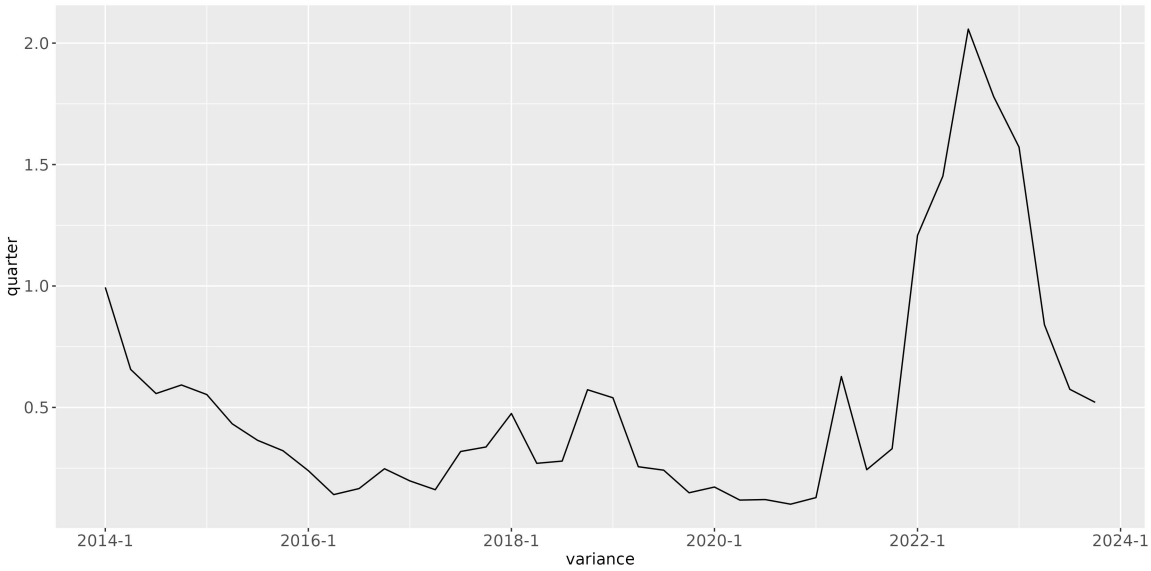


Figure 6: Cross-sectional variance of expectation by quarter

Table 6: Phillips curve estimation under various specifications

	Dependent variable:			
	inflation			
	(1)	(2)	(3)	(4)
log(marginal cost)	0.173^{***}	0.085^{***}	0.005	0.005
	(0.020)	(0.020)	(0.004)	(0.006)
expectation		0.027^{***}		0.006^{***}
		(0.002)		(0.001)
lag(inflation, 1)			0.965^{***}	0.947^{***}
			(0.007)	(0.012)
time FE	YES	YES	YES	YES
sector FE	YES	YES	YES	YES
Observations	1,080	560	1,064	560
R^2	0.074	0.333	0.939	0.949
Adjusted R^2	−0.019	0.314	0.933	0.947
F Statistic	73.644^{***}	158.713^{***}	$10{,}577.590^{***}$	$3{,}324.676^{***}$
Note:				*p<0.1; **p<0.05; ***p<0.01

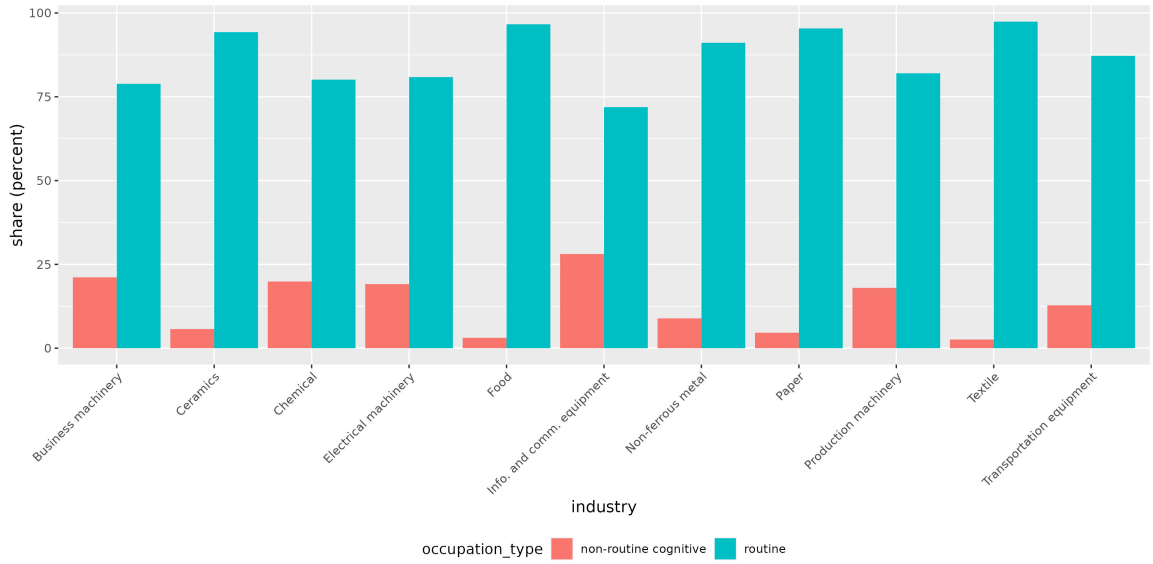


Figure 7: Occupational composition by industry

Note: non-routine manual jobs were excluded, as they only occupy a small share of employment.